

**MP2520F IO Cell Information**

MD2520FIOcell Info

| Index | Pin Name  | I/O | Alt Pin Name/Function | I/O | Alt Function 1                                  | I/O | Alt Function 2 | PAD           | RESET State | STOP State | Freq. | SSO | Cap. |
|-------|-----------|-----|-----------------------|-----|-------------------------------------------------|-----|----------------|---------------|-------------|------------|-------|-----|------|
| 1     | GPIOA[15] |     | GPIO/VD[23]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 2     | GPIOA[14] |     | GPIO/VD[22]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 3     | GPIOA[13] |     | GPIO/VD[21]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 4     | GPIOA[12] |     | GPIO/VD[20]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 5     | GPIOA[11] |     | GPIO/VD[19]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 6     | GPIOA[10] |     | GPIO/VD[18]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 7     | GPIOA[9]  |     | GPIO/VD[17]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 8     | GPIOA[8]  |     | GPIO/VD[16]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 9     | GPIOA[7]  |     | GPIO/VD[15]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 10    | GPIOA[6]  |     | GPIO/VD[14]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 11    | GPIOA[5]  |     | GPIO/VD[13]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 12    | GPIOA[4]  |     | GPIO/VD[12]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 13    | GPIOA[3]  |     | GPIO/VD[11]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 14    | GPIOA[2]  |     | GPIO/VD[10]           | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 15    | GPIOA[1]  |     | GPIO/VD[9]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 16    | GPIOA[0]  |     | GPIO/VD[8]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 17    | GPIOB[15] |     | GPIO/VD[7]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 18    | GPIOB[14] |     | GPIO/VD[6]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 19    | GPIOB[13] |     | GPIO/VD[5]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 20    | GPIOB[12] |     | GPIO/VD[4]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 21    | GPIOB[11] |     | GPIO/VD[3]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 22    | GPIOB[10] |     | GPIO/VD[2]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 23    | GPIOB[9]  |     | GPIO/VD[1]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 24    | GPIOB[8]  |     | GPIO/VD[0]            | O   | Video output data (RGB, Multiplexed-RGB, YCbCr) |     |                | phbsu100ct8sm | GPIO IN     |            | 36M   | SSO | 15pF |
| 25    | GPIOB[7]  |     | GPIO/CLKH             | O   | Pixel Clock Output                              |     |                | phbsu100ct8sm | GPIO IN     |            | 30K   |     | 15pF |
| 26    | GPIOB[6]  |     | GPIO/DE               | O   | Data Enable                                     |     |                | phbsu100ct8sm | GPIO IN     |            | 30K   |     | 15pF |
| 27    | GPIOB[5]  |     | GPIO/HSYNC            | O   | Horizontal Sync                                 |     |                | phbsu100ct8sm | GPIO IN     |            | 30K   |     | 15pF |
| 28    | GPIOB[4]  |     | GPIO/VSYSN            | O   | Vertical Sync                                   |     |                | phbsu100ct8sm | GPIO IN     |            | 30K   |     | 15pF |
| 29    | GPIOB[3]  |     | GPIO/POL              | O   | Data recerse signal of source driver            |     |                | phbsu100ct8sm | GPIO IN     |            | 30K   |     | 15pF |
| 30    | GPIOB[2]  |     | GPIO/CLKV             | O   | Vertical Clock output                           |     |                | phbsu100ct8sm | GPIO IN     |            | 30K   |     | 15pF |
| 31    | GPIOB[1]  |     | GPIO/PS               | O   | Power Savel                                     |     |                | phbsu100ct8sm | GPIO IN     |            | 30K   |     | 15pF |
| 32    | GPIOB[0]  |     | GPIO/FG               | O   | Clock signal of gate driver                     |     |                | phbsu100ct8sm | GPIO IN     |            | 30K   |     | 15pF |
| 33    | GPIOC[15] |     | GPIO/ID[15]           | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 34    | GPIOC[14] |     | GPIO/ID[14]           | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 35    | GPIOC[13] |     | GPIO/ID[13]           | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 36    | GPIOC[12] |     | GPIO/ID[12]           | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 37    | GPIOC[11] |     | GPIO/ID[11]           | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 38    | GPIOC[10] |     | GPIO/ID[10]           | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 39    | GPIOC[9]  |     | GPIO/ID[9]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 40    | GPIOC[8]  |     | GPIO/ID[8]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 41    | GPIOC[7]  |     | GPIO/ID[7]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 42    | GPIOC[6]  |     | GPIO/ID[6]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 43    | GPIOC[5]  |     | GPIO/ID[5]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 44    | GPIOC[4]  |     | GPIO/ID[4]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 45    | GPIOC[3]  |     | GPIO/ID[3]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 46    | GPIOC[2]  |     | GPIO/ID[2]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 47    | GPIOC[1]  |     | GPIO/ID[1]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 48    | GPIOC[0]  |     | GPIO/ID[0]            | I   | Image Capture Data Input                        |     |                | phbsu100ct8sm | GPIO IN     |            | 10M   | SSO | 15pF |
| 49    | GPIOD[0]  |     | GPIO/TX0              | O   | UART0 Transmit Pin                              |     |                | phbsu100ct8sm | GPIO IN     |            | 4M    |     | 15pF |
| 50    | GPIOD[1]  |     | GPIO/RX0              | I   | UART0 Receive Pin                               |     |                | phbsu100ct8sm | GPIO IN     |            | 4M    |     | 15pF |

|     |           |                      |   |                                      |                                                 |                |               |  |     |          |
|-----|-----------|----------------------|---|--------------------------------------|-------------------------------------------------|----------------|---------------|--|-----|----------|
| 51  | GPIOD[2]  | GPIO/nRIO            | I | UART1, UART2 Ring Indicator Pin      |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 52  | GPIOD[3]  | GPIO/nDCD            | I | UART1, UART2 Data-Carrier-Detect Pin |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 53  | GPIOD[4]  | GPIO/nDSR            | I | UART1, UART2 Data-Set-Ready Pin      |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 54  | GPIOD[5]  | GPIO/nDTR            | O | UART1, UART2 Data-Terminal-Ready Pin |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 55  | GPIOD[6]  | GPIO/nCTS            | I | UART1, UART2 Clear-To-Send Pin       |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 56  | GPIOD[7]  | GPIO/nRTS            | O | UART1, UART2 Ready-To-Send Pin       |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 57  | GPIOD[8]  | GPIO/TX1             | O | UART1 Transmit Pin                   |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 58  | GPIOD[9]  | GPIO/RX1             | I | UART1 Receive Pin                    |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 59  | GPIOD[10] | GPIO/RX2             | I | UART2/IRRX Receive Pin               |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 60  | GPIOD[11] | GPIO/TX2             | O | UART2/IRTX Transmit Pin              |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 61  | GPIOD[12] | GPIO/RX3/SSPEXTCLKIN | I | UART3 Receive Pin                    | Synchronous Serial Port<br>external clock input | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 62  | GPIOD[13] | GPIO/TX3             | O | UART3 Transmit Pin                   |                                                 | phbsu100ct8sm  | GPIO IN       |  | 4M  | 15pF     |
| 63  | GPIOE[15] | GPIO/ZD[15]          | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 64  | GPIOE[14] | GPIO/ZD[14]          | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 65  | GPIOE[13] | GPIO/ZD[13]          | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 66  | GPIOE[12] | GPIO/ZD[12]          | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 67  | GPIOE[11] | GPIO/ZD[11]          | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 68  | GPIOE[10] | GPIO/ZD[10]          | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 69  | GPIOE[9]  | GPIO/ZD[9]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 70  | GPIOE[8]  | GPIO/ZD[8]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 71  | GPIOE[7]  | GPIO/ZD[7]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 72  | GPIOE[6]  | GPIO/ZD[6]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 73  | GPIOE[5]  | GPIO/ZD[5]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 74  | GPIOE[4]  | GPIO/ZD[4]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 75  | GPIOE[3]  | GPIO/ZD[3]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 76  | GPIOE[2]  | GPIO/ZD[2]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 77  | GPIOE[1]  | GPIO/ZD[1]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 78  | GPIOE[0]  | GPIO/ZD[0]           | B | Non DRAM Data Bus                    |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 79  | GPIOF[9]  | GPIO/ZA[25]/OM[3]    | O | Non DRAM Address Bus                 | Test Mode Pin 3                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 80  | GPIOF[8]  | GPIO/ZA[24]/OM[2]    | O | Non DRAM Address Bus                 | Test Mode Pin 2                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 81  | GPIOF[7]  | GPIO/ZA[23]/OM[1]    | O | Non DRAM Address Bus                 | Test Mode Pin 1                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 82  | GPIOF[6]  | GPIO/ZA[22]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 83  | GPIOF[5]  | GPIO/ZA[21]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 84  | GPIOF[4]  | GPIO/ZA[20]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 85  | GPIOF[3]  | GPIO/ZA[19]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 86  | GPIOF[2]  | GPIO/ZA[18]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 87  | GPIOF[1]  | GPIO/ZA[17]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 88  | GPIOF[0]  | GPIO/ZA[16]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 89  | GPIOG[15] | GPIO/ZA[15]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 90  | GPIOG[14] | GPIO/ZA[14]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 91  | GPIOG[13] | GPIO/ZA[13]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 92  | GPIOG[12] | GPIO/ZA[12]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 93  | GPIOG[11] | GPIO/ZA[11]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 94  | GPIOG[10] | GPIO/ZA[10]          | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 95  | GPIOG[9]  | GPIO/ZA[9]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 96  | GPIOG[8]  | GPIO/ZA[8]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 97  | GPIOG[7]  | GPIO/ZA[7]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 98  | GPIOG[6]  | GPIO/ZA[6]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 99  | GPIOG[5]  | GPIO/ZA[5]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 100 | GPIOG[4]  | GPIO/ZA[4]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 101 | GPIOG[3]  | GPIO/ZA[3]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 102 | GPIOG[2]  | GPIO/ZA[2]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 103 | GPIOG[1]  | GPIO/ZA[1]           | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |
| 104 | GPIOG[0]  | GPIO/ZA[0](DQMz[0])  | O | Non DRAM Address Bus                 |                                                 | phbsu100ct12sm | Alt Function1 |  | 50M | SSO 15pF |

|     |           |                        |   |                                                                                                                                                                                                  |                      |                |               |  |     |  |      |
|-----|-----------|------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|---------------|--|-----|--|------|
| 105 | GPIOH[6]  | GPIO/XDOFF             | O | Gate-OFF Control signal fo gate driver                                                                                                                                                           |                      | phbsu100ct8sm  | GPIO IN       |  | 30K |  | 15pF |
| 106 | GPIOH[5]  | GPIO/VCLKIN            | I | Video Clock Input                                                                                                                                                                                |                      | phbsu100ct8sm  | GPIO IN       |  | 30K |  | 15pF |
| 107 | GPIOH[4]  | GPIO/ICLKIN            | I | Image Clock In                                                                                                                                                                                   |                      | phbsu100ct8sm  | GPIO IN       |  | 30K |  | 15pF |
| 108 | GPIOH[3]  | GPIO/IHSYNC            | B | Image HSYNC                                                                                                                                                                                      |                      | phbsu100ct8sm  | GPIO IN       |  | 30K |  | 15pF |
| 109 | GPIOH[2]  | GPIO/IVSYNC            | B | Image VSYNC                                                                                                                                                                                      |                      | phbsu100ct8sm  | GPIO IN       |  | 30K |  | 15pF |
| 110 | GPIOH[1]  | GPIO/ICLKOUT           | O | Image Clock Out                                                                                                                                                                                  |                      | phbsu100ct8sm  | GPIO IN       |  | 30K |  | 15pF |
| 111 | GPIOH[0]  | GPIO/EXTCLKO           | O | External Clock Out                                                                                                                                                                               |                      | phbsu100ct8sm  | Alt Function1 |  | 50M |  | 15pF |
| 112 | GPIOJ[15] | CBBUSREQ               | I | CardBus BUS Request/V656IN[7]                                                                                                                                                                    |                      | phbsu100ct8sm  | Alt Function1 |  | 50M |  |      |
| 113 | GPIOJ[14] | CBBUSGNT               | O | CardBus BUS Grant/V656IN[6]                                                                                                                                                                      |                      | phbsu100ct8sm  | Alt Function1 |  | 50M |  |      |
| 114 | GPIOJ[13] | GPIO/nPWAIT            | I | Wait Control for PCMCIA/CF/IDE/Static Memory. This signal is an input and is driven low by the PCMCIA/CF/IDE/Static Memory to extend the length of the transfers to/from applications processor. |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 115 | GPIOJ[12] | GPIO/DQMz[1]           | O | Static Memory Data Mask[1], Byte Enable, = nByteEnb                                                                                                                                              |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 116 | GPIOJ[11] | GPIO/nSCS[3]/656IN[7]  | O | Static Memory Chip Select                                                                                                                                                                        | 656 type video input | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 117 | GPIOJ[10] | GPIO/nSCS[2]/656IN[6]  | O | Static Memory Chip Select                                                                                                                                                                        | 656 type video input | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 118 | GPIOJ[9]  | GPIO/nSCS[1]/656IN[5]  | O | Static Memory Chip Select                                                                                                                                                                        | 656 type video input | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 119 | GPIOJ[8]  | GPIO/nSCS[0]           | O | Static Memory Chip Select                                                                                                                                                                        |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 120 | GPIOJ[7]  | GPIO/nPCS[1]           | O | PCMCIA/CF Chip Select                                                                                                                                                                            |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 121 | GPIOJ[6]  | GPIO/nPCS[0]           | O | PCMCIA/CF Chip Select                                                                                                                                                                            |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 122 | GPIOJ[5]  | GPIO/nICS[1]           | O | IDE Chip Select                                                                                                                                                                                  |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 123 | GPIOJ[4]  | GPIO/nICS[0]           | O | IDE Chip Select                                                                                                                                                                                  |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 124 | GPIOJ[3]  | GPIO/nNFCE[3]/656IN[4] | O | NAND Chip Select/V656IN[5]                                                                                                                                                                       | 656 type video input | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 125 | GPIOJ[2]  | GPIO/nNFCE[2]/656IN[3] | O | NAND Chip Select/V656IN[4]                                                                                                                                                                       | 656 type video input | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 126 | GPIOJ[1]  | GPIO/nNFCE[1]/656IN[2] | O | NAND Chip Select/V656IN[3]                                                                                                                                                                       | 656 type video input | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 127 | GPIOJ[0]  | GPIO/nNFCE[0]          | O | NAND Chip Select                                                                                                                                                                                 |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 128 | GPIOJ[15] | CBREQ                  | I | CardBus Request/V656IN[2]                                                                                                                                                                        |                      | phbsu100ct12sm | Alt Function1 |  |     |  |      |
| 129 | GPIOJ[14] | CBACK                  | I | CardBus Ack/V656IN[1]                                                                                                                                                                            |                      | phbsu100ct12sm | Alt Function1 |  |     |  |      |
| 130 | GPIOJ[13] | CBRDY                  | I | CardBus Ready/V656IN[0]                                                                                                                                                                          |                      | phbsu100ct12sm | Alt Function1 |  |     |  |      |
| 131 | GPIOJ[12] | GPIO/nPOE              | O | PCMCIA/CF/SM Output Enable. This signal is an output and performs read from memory and attribute space.                                                                                          |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 132 | GPIOJ[11] | GPIO/nPWE/WRITE_CRB    | O | PCMCIA/CF/SM Write Enable. This signal is an output and performs writes to memory and attribute space.                                                                                           |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 133 | GPIOJ[10] | GPIO/nPIOR             | O | PCMCIA/CF/IDE I/O Read Enable. This signal is an output and performs read transactions from the PCMCIA I/O space.                                                                                |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 134 | GPIOJ[9]  | GPIO/nPIOW             | O | PCMCIA/CF/IDE I/O Write Enable. This signal is an output and performs write transactions to the PCMCIA I/O space.                                                                                |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 135 | GPIOJ[8]  | GPIO/PSKSEL            | O | PCMCIA/CF Socket Select. This signal is an output and is used by external steering logic to route control, address and data signals to one of the two PCMCIA sockets.                            |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |
| 136 | GPIOJ[7]  | GPIO/nPREG             | O | PCMCIA/CF Register Select. This signal is an output and indicates that, on a memory transaction, the target address is attribute space. This signal has the same timing as address               |                      | phbsu100ct12sm | Alt Function1 |  | 50M |  | 15pF |

|     |           |                             |   |                                                                                                                                                                   |   |                                    |                |               |  |      |     |      |
|-----|-----------|-----------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------|----------------|---------------|--|------|-----|------|
| 137 | GPIOJ[6]  | GPIO/nPIOIS16               | I | PCMCIA/CF/IDE Select 16bit Access. This signal is an input and is an acknowledge from the PCMCIA card that the current address is a valid 16 bit wide I/O address |   |                                    | phbsu100ct12sm | Alt Function1 |  | 50M  |     | 15pF |
| 138 | GPIOJ[5]  | GPIO/RDnWR                  | O | Buffer Direction Control. Read/Write for static interface. Intended for use as steering signal for buffering logic.                                               |   |                                    | phbsu100ct12sm | Alt Function1 |  | 50M  |     | 15pF |
| 139 | GPIOJ[4]  | GPIO/nBUFOE                 | O | Buffer Output Enable                                                                                                                                              |   |                                    | phbsu100ct12sm | Alt Function1 |  | 50M  |     | 15pF |
| 140 | GPIOJ[3]  | GPIO/nBUFENB                | O | Buffer Enable                                                                                                                                                     |   |                                    |                |               |  |      |     |      |
| 141 | GPIOJ[2]  | GPIO/RnB                    | I | NAND Ready & Busy                                                                                                                                                 |   |                                    | phbsu100ct12sm | Alt Function1 |  | 50M  |     | 15pF |
| 142 | GPIOJ[1]  | GPIO/nNFOE                  | O | NAND Output Enable                                                                                                                                                |   |                                    | phbsu100ct12sm | Alt Function1 |  | 50M  |     | 15pF |
| 143 | GPIOJ[0]  | GPIO/nNFW                   | O | NAND Write Enable                                                                                                                                                 |   |                                    | phbsu100ct12sm | Alt Function1 |  | 50M  |     | 15pF |
| 144 | GPIOK[7]  | GPIO/DREQ3/STVD             | I | DMA Request                                                                                                                                                       | B | Shift clock output for Gate        | phbsu100ct8sm  | GPIO IN       |  | 50M  |     | 15pF |
| 145 | GPIOK[6]  | GPIO/DREQ2                  | I | DMA Request                                                                                                                                                       |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 50M  |     | 15pF |
| 146 | GPIOK[5]  | GPIO/DREQ1                  | I | DMA Request                                                                                                                                                       |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 50M  |     | 15pF |
| 147 | GPIOK[4]  | GPIO/DREQ0                  | I | DMA Request                                                                                                                                                       |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 50M  |     | 15pF |
| 148 | GPIOK[3]  | GPIO/DVAL3/STHR             | O | DMA Acknowledge                                                                                                                                                   | B | Start Pulse of source Drive        | phbsu100ct8sm  | GPIO IN       |  | 50M  |     | 15pF |
| 149 | GPIOK[2]  | GPIO/DVAL2/V656CLK          | O | DMA Acknowledge                                                                                                                                                   | I | 656 type video input clock         | phbsu100ct8sm  | GPIO IN       |  | 50M  |     | 15pF |
| 150 | GPIOK[1]  | GPIO/DVAL1                  | O | DMA Acknowledge                                                                                                                                                   |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 50M  |     | 15pF |
| 151 | GPIOK[0]  | GPIO/DVAL0                  | O | DMA Acknowledge                                                                                                                                                   |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 50M  |     | 15pF |
| 152 | GPIOL[14] | GPIO/PWMOUT3/SSPIN          | O | Pulse Width Modulation channel 3                                                                                                                                  | I | SSP Data Input                     | phbsu100ct12sm | GPIO IN       |  | 100M |     | 15pF |
| 153 | GPIOL[13] | GPIO/PWMOUT2/PPM            | O | Pulse Width Modulation channel 2                                                                                                                                  | I | PPM Input                          | phbsu100ct12sm | GPIO IN       |  | 100M |     | 15pF |
| 154 | GPIOL[12] | GPIO/PWMOUT1                | O | Pulse Width Modulation channel 1                                                                                                                                  |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 1M   |     | 15pF |
| 155 | GPIOL[11] | GPIO/PWMOUT0                | O | Pulse Width Modulation channel 0                                                                                                                                  |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 1M   |     | 15pF |
| 156 | GPIOL[10] | GPIO/ABITCLK/I2SCLK         | I | AC97 Audio Port bit clock                                                                                                                                         | O | I2S Clock(QCLK, I2S)               | phbsu100ct8sm  | GPIO IN       |  | 5M   |     | 15pF |
| 157 | GPIOL[9]  | GPIO/ADATIN/I2SDATIN        | I | AC97 Audio Port data in                                                                                                                                           | I | I2S SDATA IN                       | phbsu100ct8sm  | GPIO IN       |  | 5M   |     | 15pF |
| 158 | GPIOL[8]  | GPIO/ASYNC/I2SSYNC          | O | AC97 Audio Port sync signal                                                                                                                                       | O | I2S SYNC                           | phbsu100ct8sm  | GPIO IN       |  | 5M   |     | 15pF |
| 159 | GPIOL[7]  | GPIO/nARST/I2SACLK          | O | AC97 Audio Port reset signal                                                                                                                                      | O | I2S System clock(ACLK)             | phbsu100ct8sm  | GPIO IN       |  | 5M   |     | 15pF |
| 160 | GPIOL[6]  | GPIO/ADATOUT/I2SDATAOUT     | O | AC97 Audio Port data out                                                                                                                                          | O | I2S DATA OUT                       | phbsu100ct8sm  | GPIO IN       |  | 5M   |     | 15pF |
| 161 | GPIOL[5]  | GPIO/SDICLK                 | O | MMC/SD Clock                                                                                                                                                      |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 30M  |     | 15pF |
| 162 | GPIOL[4]  | GPIO/SDICMD                 | B | MMC/SD Command                                                                                                                                                    |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 30M  |     | 15pF |
| 163 | GPIOL[3]  | GPIO/SDIDAT[3]/MSBS         | B | MMC/SD Data                                                                                                                                                       | O | Memory Stick BS                    | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 164 | GPIOL[2]  | GPIO/SDIDAT[2]/MSDAT        | B | MMC/SD Data                                                                                                                                                       | B | Memory Stick SDAT                  | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 165 | GPIOL[1]  | GPIO/SDIDAT[1]              | B | MMC/SD Data                                                                                                                                                       |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 166 | GPIOL[0]  | GPIO/SDIDAT[0]              | B | MMC/SD Data                                                                                                                                                       |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 167 | GPIOM[8]  | GPIO/VD656[7]/TSISYNC/LRCK  | B | 656 Type Video Data                                                                                                                                               | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 168 | GPIOM[7]  | GPIO/VD656[6]/TSIDP/BCK     | B | 656 Type Video Data                                                                                                                                               | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 169 | GPIOM[6]  | GPIO/VD656[5]/TSIERR/SDAT   | B | 656 Type Video Data                                                                                                                                               | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 170 | GPIOM[5]  | GPIO/VD656[4]/TSDI0         | B | 656 Type Video Data                                                                                                                                               | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 171 | GPIOM[4]  | GPIO/VD656[3]/TSDICLK       | B | 656 Type Video Data                                                                                                                                               |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 172 | GPIOM[3]  | GPIO/VD656[2]               | B | 656 Type Video Data                                                                                                                                               | I | CD I/F SDATA, MPEG I/F Serial Data | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 173 | GPIOM[2]  | GPIO/VD656[1]               | B | 656 Type Video Data                                                                                                                                               | I | CD I/F LRCK                        | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 174 | GPIOM[1]  | GPIO/VD656[0]               | B | 656 Type Video Data                                                                                                                                               | I | CD I/F BCLK                        | phbsu100ct8sm  | GPIO IN       |  | 30M  | SSO | 15pF |
| 175 | GPIOM[0]  | GPIO/VDCLKIN/TSICLK         | B | 656 Type Video Clock                                                                                                                                              | I | MPEG I/F Clock                     | phbsu100ct8sm  | GPIO IN       |  | 30M  |     | 15pF |
| 176 | GPION[7]  | GPIO/TSIDJ[7]/TSISYNC(CDRX) | I | MPEG Interface Data In                                                                                                                                            | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 10M  | SSO | 15pF |
| 177 | GPION[6]  | GPIO/TSIDJ[6]/TSIDP(CDTX)   | I | MPEG Interface Data In                                                                                                                                            | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 10M  | SSO | 15pF |
| 178 | GPION[5]  | GPIO/TSIDJ[5]/TSIERR(LRCK)  | I | MPEG Interface Data In                                                                                                                                            | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 10M  | SSO | 15pF |
| 179 | GPION[4]  | GPIO/TSIDJ[4]/TSDI0(BCK)    | I | MPEG Interface Data In                                                                                                                                            | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 10M  | SSO | 15pF |
| 180 | GPION[3]  | GPIO/TSIDJ[3]/TSICLK(SDAT)  | I | MPEG Interface Data In                                                                                                                                            | I | MPEG I/F Control In                | phbsu100ct8sm  | GPIO IN       |  | 10M  | SSO | 15pF |
| 181 | GPION[2]  | GPIO/TSIDJ[2]/MSINS         | I | MPEG Interface Data In                                                                                                                                            | I | Memory Stick INS                   | phbsu100ct8sm  | GPIO IN       |  | 10M  | SSO | 15pF |
| 182 | GPION[1]  | GPIO/TSIDJ[1]/MSCLK         | I | MPEG Interface Data In                                                                                                                                            | O | Memory Stick Clock                 | phbsu100ct8sm  | GPIO IN       |  | 10M  | SSO | 15pF |
| 183 | GPION[0]  | GPIO/TSIDJ[0]               | I | MPEG Interface Data In                                                                                                                                            |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 10M  | SSO | 15pF |
| 184 | GPIOO[5]  | GPIO/SPDIFIN                | I | SPDIF In                                                                                                                                                          |   |                                    | phbsu100ct8sm  | GPIO IN       |  | 4M   |     | 15pF |

|     |            |   |                                                                                                                                                                                                                                                                                                                    |   |                                      |  |  |                |         |  |       |     |      |
|-----|------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------|--|--|----------------|---------|--|-------|-----|------|
| 185 | GPIO[4]    |   | GPIO/SPDIFOUT                                                                                                                                                                                                                                                                                                      | O | SPDIF Out                            |  |  | phbsu100ct8sm  | GPIO IN |  | 4M    |     | 15pF |
| 186 | GPIO[3]    |   | GPIO/SSPCLK                                                                                                                                                                                                                                                                                                        | O | Synchronous Serial Port Clock        |  |  | phbsu100ct8sm  | GPIO IN |  | 4M    |     | 15pF |
| 187 | GPIO[2]    |   | GPIO/SSPFRM                                                                                                                                                                                                                                                                                                        | O | Synchronous Serial Port Frame Signal |  |  | phbsu100ct8sm  | GPIO IN |  | 4M    |     | 15pF |
| 188 | GPIO[1]    |   | GPIO/SSPDAT                                                                                                                                                                                                                                                                                                        | B | Synchronous Serial Port Data         |  |  | phbsu100ct8sm  | GPIO IN |  | 4M    |     | 15pF |
| 189 | GPIO[0]    |   | GPIO/1WIRE                                                                                                                                                                                                                                                                                                         | O | 1 wire Master Output                 |  |  | phbsd8sm       | GPIO IN |  | 1M    |     | 15pF |
| 190 | OM[0]      | I | Test Mode (External Clock selection)                                                                                                                                                                                                                                                                               |   |                                      |  |  | phis           | Normal  |  | 1     |     | 15pF |
| 191 | TEST_EN    | I | Test Mode Enable                                                                                                                                                                                                                                                                                                   |   |                                      |  |  | phis           | Normal  |  | 1     |     | 15pF |
| 192 | nPORSEL    | I | POWER On Reset Selection. This signal determines whether the internal POR circuit to be used or nRESET to be used. If nPORSEL is 0, internal POR circuit would be used.                                                                                                                                            |   |                                      |  |  | phia_abb       | Normal  |  | 1     |     | 15pF |
| 193 | nRESET     | I | Global Reset In. Active low input. nRESET is a level sensitive input which is used to start the processor from a known address. A LOW level will cause the current instruction to terminate abnormally, and all on-chip state to be reset. When nRESET is driven HIGH, the processor will re-start from address 0. |   |                                      |  |  | phis           | Normal  |  | 1     |     | 15pF |
| 194 | nGRESETOut | O | Global Reset Out                                                                                                                                                                                                                                                                                                   |   |                                      |  |  | phob8sm        | Normal  |  | 1     |     | 15pF |
| 195 | nPWRRGLTOn | O | Active Low Power Control                                                                                                                                                                                                                                                                                           |   |                                      |  |  | phob8          | Normal  |  | 1     |     | 15pF |
| 196 | EXTCLK     | I | External Clock In                                                                                                                                                                                                                                                                                                  |   |                                      |  |  | phis           | Normal  |  | 50M   |     | 15pF |
| 197 | XTI        | I | 7Mhz X-tal In                                                                                                                                                                                                                                                                                                      |   |                                      |  |  | phsosc16       | Normal  |  | 7M    |     | 15pF |
| 198 | XTO        | O | 7Mhz X-tal Out                                                                                                                                                                                                                                                                                                     |   |                                      |  |  |                | Normal  |  | 7M    |     | 15pF |
| 199 | TCK        | I | Clock input for the JTAG Logic                                                                                                                                                                                                                                                                                     |   |                                      |  |  | phis           | Normal  |  | 20M   |     | 15pF |
| 200 | nTRST      | I | Reset input for the JTAG Logic                                                                                                                                                                                                                                                                                     |   |                                      |  |  | phis           | Normal  |  | 1     |     | 15pF |
| 201 | TDO        | O | Serial output for test instructions and data for JTAG                                                                                                                                                                                                                                                              |   |                                      |  |  | phot8          | Normal  |  | 20M   |     | 15pF |
| 202 | TMS        | I | TMS controls the sequence of the TAP controller's state                                                                                                                                                                                                                                                            |   |                                      |  |  | phis           | Normal  |  | 20M   |     | 15pF |
| 203 | TDI        | I | Serial Data input for JTAG                                                                                                                                                                                                                                                                                         |   |                                      |  |  | phis           | Normal  |  | 20M   |     | 15pF |
| 204 | XTIRTC     | I | RTC Crystal input                                                                                                                                                                                                                                                                                                  |   |                                      |  |  | phgpad_option  | Normal  |  | 32768 |     | 15pF |
| 205 | XTORTC     | O | RTC Crystal output                                                                                                                                                                                                                                                                                                 |   |                                      |  |  | phgpad_option  | Normal  |  | 32768 |     | 15pF |
| 206 | EXTIN0     | I | Wake-up Control input #0                                                                                                                                                                                                                                                                                           |   |                                      |  |  | phis           | Normal  |  | 1     |     | 15pF |
| 207 | EXTIN1     | I | Wake-up Control Input #1                                                                                                                                                                                                                                                                                           |   |                                      |  |  | phis           | Normal  |  | 1     |     | 15pF |
| 208 | nBATTF     | I | Battery Fault. Active low input. Signals MMSP2 that the main power source is going away (battery is low or is removed from the system.)                                                                                                                                                                            |   |                                      |  |  | phis           | Normal  |  | 1     |     | 15pF |
| 209 | SDCLK_X    | O | SDRAM clock                                                                                                                                                                                                                                                                                                        |   |                                      |  |  | phob16sm       | Normal  |  | 100M  |     | 15pF |
| 210 | XA[14:0]   | O | SDRAM address bus. This bus signals the address requested for memory accesses.                                                                                                                                                                                                                                     |   |                                      |  |  | phob8          | Normal  |  | 50M   | SSO | 15pF |
| 225 | XD[15:0]   | B | SDRAM data bus. XD[15:0] are used for 16-bit data mode.                                                                                                                                                                                                                                                            |   |                                      |  |  | phbsu100ct12sm | Normal  |  | 100M  | SSO | 15pF |
| 226 | XD[31:16]  | B | SDRAM data bus. These data bits are used for 32-bit memories.                                                                                                                                                                                                                                                      |   |                                      |  |  | phbsu100ct12sm | Normal  |  | 100M  | SSO | 15pF |
| 257 | SCKEx      | O | SDRAM clock enable.                                                                                                                                                                                                                                                                                                |   |                                      |  |  | phob8          | Normal  |  | 50M   |     | 15pF |
| 258 | nSCSx0     | O | SDRAM CS for banks 0. This signal should be connected to the chip select (CS) pins for SDRAM.                                                                                                                                                                                                                      |   |                                      |  |  | phob8          | Normal  |  | 50M   |     | 15pF |

|     |            |   |                                                                                                                                                                                            |  |  |  |          |        |  |     |  |      |
|-----|------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|----------|--------|--|-----|--|------|
| 259 | nSCSx1     | O | SDRAM CS for banks 1. This signal should be connected to the chip select (CS) pins for SDRAM.                                                                                              |  |  |  | phob8    | Normal |  | 50M |  | 15pF |
| 260 | nSRASx     | O | SDRAM RAS. This signal should be connected to the row address strobe(RAS) pins for all banks of SDRAM.                                                                                     |  |  |  | phob8    | Normal |  | 50M |  | 15pF |
| 261 | nSCASx     | O | SDRAM CAS. This signal should be connected to the column address strobe (CAS) pins for all banks of SDRAM.                                                                                 |  |  |  | phob8    | Normal |  | 50M |  | 15pF |
| 262 | nSWEx      | O | SDRAM write enable. This signal should be connected to the write enables of SDRAM.                                                                                                         |  |  |  | phob8    | Normal |  | 50M |  | 15pF |
| 263 | SDQMx[3:0] | O | SDRAM DQM for data bytes 0 through 3. These signals should be connected to the data output mask enables (DQM) for SDRAM                                                                    |  |  |  | phob8    | Normal |  | 50M |  | 15pF |
| 267 | ADCIN0     | I | ADC Analog Input                                                                                                                                                                           |  |  |  | phia_abb | Normal |  | 2M  |  | 15pF |
| 268 | ADCIN1     | I | ADC Analog Input                                                                                                                                                                           |  |  |  | phia_abb | Normal |  | 2M  |  | 15pF |
| 269 | ADCIN2     | I | ADC Analog Input                                                                                                                                                                           |  |  |  | phia_abb | Normal |  | 2M  |  | 15pF |
| 270 | ADCIN3     | I | ADC Analog Input                                                                                                                                                                           |  |  |  | phia_abb | Normal |  | 2M  |  | 15pF |
| 271 | ADCIN4     | I | ADC Analog Input                                                                                                                                                                           |  |  |  | phia_abb | Normal |  | 2M  |  | 15pF |
| 272 | ADCIN5     | I | ADC Analog Input                                                                                                                                                                           |  |  |  | phia_abb | Normal |  | 2M  |  | 15pF |
| 273 | VREF       | I | ADC Voltage Reference                                                                                                                                                                      |  |  |  | phia_abb | Normal |  | 1   |  | 15pF |
| 274 | DP[0]      | B | USB Host/USB Device port positive Pin of differential pair                                                                                                                                 |  |  |  | pbusb1   | Normal |  | 12M |  | 15pF |
| 275 | DN[0]      | B | USB Host/USB Device port negative Pin of differential pair                                                                                                                                 |  |  |  | pbusb1   | Normal |  | 12M |  | 15pF |
| 276 | DP[1]      | B | USB Host port positive Pin of differential pair                                                                                                                                            |  |  |  | pbusb1   | Normal |  | 12M |  | 15pF |
| 277 | DN[1]      | B | USB Host port negative Pin of differential pair                                                                                                                                            |  |  |  | pbusb1   | Normal |  | 12M |  | 15pF |
| 278 | DP[2]      | B | USB Host/USB Device port positive Pin of differential pair                                                                                                                                 |  |  |  | pbusb1   | Normal |  | 12M |  | 15pF |
| 279 | DN[2]      | B | USB Host/USB Device port negative Pin of differential pair                                                                                                                                 |  |  |  | pbusb1   | Normal |  | 12M |  | 15pF |
| 280 | SCL        | B | I2C Clock. This signal is bidirectional. When it is driving, it functions as an open collector device and requires a pull up resistor. As an input, it expects standard CMOS levels.       |  |  |  | phbsd8sm | Normal |  | 1M  |  | 15pF |
| 281 | SDA        | B | I2C Data signal. This signal is bidirectional. When it is driving, it functions as an open collector device and requires a pull up resistor. As an input, it expects standard CMOS levels. |  |  |  | phbsd8sm | Normal |  | 1M  |  | 15pF |
|     | VDDUPLL    | P |                                                                                                                                                                                            |  |  |  |          |        |  |     |  |      |
|     | VSSUPLL    | G |                                                                                                                                                                                            |  |  |  |          |        |  |     |  |      |
|     | VDDFPLL    | P |                                                                                                                                                                                            |  |  |  |          |        |  |     |  |      |
|     | VSSFPLL    | G |                                                                                                                                                                                            |  |  |  |          |        |  |     |  |      |
|     | VDDAPLL    | P |                                                                                                                                                                                            |  |  |  |          |        |  |     |  |      |
|     | VSSAPLL    | G |                                                                                                                                                                                            |  |  |  |          |        |  |     |  |      |

|           |   |  |  |  |  |  |  |  |  |  |  |  |  |
|-----------|---|--|--|--|--|--|--|--|--|--|--|--|--|
| VDDRTC    | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSRTC    | G |  |  |  |  |  |  |  |  |  |  |  |  |
| VDDADC    | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSADC    | G |  |  |  |  |  |  |  |  |  |  |  |  |
| VDDi      | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSi      | G |  |  |  |  |  |  |  |  |  |  |  |  |
| VDDa920   | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSa920   | G |  |  |  |  |  |  |  |  |  |  |  |  |
| VDDa940   | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSa940   | G |  |  |  |  |  |  |  |  |  |  |  |  |
| VDDOP     | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSOP     | G |  |  |  |  |  |  |  |  |  |  |  |  |
| VDDOPalve | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSOPalve | G |  |  |  |  |  |  |  |  |  |  |  |  |
| VDDalve1  | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSalve1  | G |  |  |  |  |  |  |  |  |  |  |  |  |
| VDDalve2  | P |  |  |  |  |  |  |  |  |  |  |  |  |
| VSSalve2  | G |  |  |  |  |  |  |  |  |  |  |  |  |

**Hardware Config Pins.**

|                  |                                |                                                      |
|------------------|--------------------------------|------------------------------------------------------|
| GPIOA[14]        | : RstCfgNFBoot                 | Nand Boot Enable                                     |
| GPIOA[13]        | : RstCfgNFBSize                | Nand Boot Size : 0 : 2048byte, 1 : 4096byte          |
| GPIOA[12]        | : RstCfgNFBUSWidth             | Nand flash Bus Width : 0 : 8bit, 1 : 16bit           |
| GPIOA[11]        | : RstCfgNFType                 | Nand flagh Type : 0 : 3 Address, 1 : 4 Address       |
| GPIOA[10]        | : RstCfgSR0BUSWidth            | SRAM Bus Width : 0 : 8bit, 1 : 16 bit                |
| GPIOA[ 9]        | : RstCfgSDRTYPE                | SDRAM Type 0 : SDRAM, 1 : DDR                        |
| GPIOA[ 8]        | : RstCfgBUSCBW                 | SDRAM Total Bus Width : 0 16 bit 1 : 32 bit          |
| GPIOA[ 7]        | : RstCfgSDRBW[1]               | SDRAM Bus Width                                      |
| GPIOA[ 6]        | : RstCfgSDRBW[0]               | 00 : 4bit, 01 : 8bit, 10 : 16bit                     |
| GPIOA[ 5]        | : RstCfgSDRCAP[1]              | SDRAM Capacity                                       |
| GPIOA[ 4]        | : RstCfgSDRCAP[0]              | 00 : 64bit, 01 : 128 bit, 10 : 256 bit, 11 : 512 bit |
| GPIOA[ 3]        | : RstCfgShadow                 | SDRAM Shandow Enable                                 |
|                  |                                |                                                      |
| GPIOA[ 1]        | : RstCfgUARTBootClk            | UART Boot Clock : 0 : PLL clock, 1 : ABITCLK         |
| GPIOA[ 0]        | : RstCfgBootDown               | UART Boot Enable                                     |
|                  |                                |                                                      |
| GPIOB[15]        | : ARM940T DEBUG Enable         | ARM940 ICE Debug Enable                              |
| GPIOB[14]        | : RstCfgSRBuf                  | SRAM Buffer Enable                                   |
| GPIOB[13]        | : RstCfgNFBuf                  | Nand flash Buffer Enbale                             |
|                  |                                |                                                      |
| TEST Mode Signal |                                |                                                      |
| TEST_EN          | Test Enable                    | 0 : Normal mode 1 : Test mode                        |
| OM0              | External Clock Enable for FCLK | 0 : PLL clock enable 1 : External Clock enable       |
| GPIOF[9]         | OM[3]                          | if TEST_EN is high this signals are used test pins.  |
| GPIOF[8]         | OM[2]                          |                                                      |
| GPIOF[7]         | OM[1]                          |                                                      |